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# CHAPTER 13 — BEST PRACTICES FOR MMU DESIGN

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***A maximum-detail, RTOS/QNX-style technical breakdown of recommended MMU design patterns, safe configuration rules, performance optimization strategies, and architectural guidelines for robust ARMv8 memory systems.***

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## 13.1 — Introduction to MMU Best Practices

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Designing a correct, safe, and high-performance MMU configuration requires:

- correct descriptor layout
- correct TCR/MAIR configuration

- correct TTBR usage
- correct ASID/VMID allocation
- correct barrier usage
- correct BBM sequences
- correct TLB invalidation
- correct device memory handling
- correct SMP ordering
- correct virtualization behavior

This chapter consolidates all prior chapters into a **practical design guide**.

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# 13.2 — Best Practices for TTBR Usage

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### TTBR0_EL1 (User Space)

- Use per-process TTBR0.
- Always assign unique ASIDs.
- Never reuse ASIDs without TLBI.
- Keep TTBR0 mappings minimal to reduce TLB pressure.

### TTBR1_EL1 (Kernel Space)

- Map kernel memory globally.
- Avoid frequent TTBR1 changes.
- Keep kernel mappings stable across boots.
- Place kernel at high VA addresses for ASLR.

### TTBR0_EL2 (Stage-2)

- Use deterministic IPA → PA mapping.
- Keep Stage-2 tables compact.
- Avoid fragmentation in Stage-2 tables.

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# 13.3 — Best Practices for TCR Configuration

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### Granule Size

- Use **4 KB granule** unless hardware requires otherwise.
- Avoid mixing granule sizes across TTBR0/TTBR1.

### VA Size

- Use **48-bit VA** for most systems.
- Avoid 52-bit VA unless absolutely necessary.

### Shareability

- Use **Inner Shareable** for SMP systems.
- Avoid Non-shareable unless mapping private memory.

### Cacheability

- Use **Write-Back** for normal memory.
- Use **Non-cacheable** only for device memory.

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# 13.4 — Best Practices for MAIR Configuration

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### Normal Memory

- AttrIndx 0 → Normal WB (0xFF)
- AttrIndx 1 → Normal NC (0x44)

### Device Memory

- AttrIndx 2 → Device-nGnRnE (0x00)
- AttrIndx 3 → Device-nGnRE (0x04)

### Rules

- Never use cacheable attributes for device memory.
- Never use device attributes for normal memory.
- Keep MAIR entries consistent across TTBR0/TTBR1.

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# 13.5 — Best Practices for Descriptor Layout

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### Alignment

- L1 block → 1 GB aligned
- L2 block → 2 MB aligned
- L3 page → 4 KB aligned
- Table pointers → 4 KB aligned

### Descriptor Types

- Use blocks for large contiguous regions.
- Use pages for fine-grained permissions.
- Avoid mixing block/page mappings in the same region.

### Permissions

- Enforce W^X (Write-xor-Execute).
- Use PXN/UXN aggressively.
- Use AP to enforce user/kernel separation.

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# 13.6 — Best Practices for ASID Allocation

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### General Rules

- Use generation-based ASID allocation.
- Avoid simple incrementing ASIDs in long-running systems.
- Never reuse ASIDs without TLBI.
- Keep kernel mappings global.

### QNX/RTOS

- Use static ASID ranges for partitions.
- Use deterministic ASID allocation for safety-critical systems.

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# 13.7 — Best Practices for VMID Allocation

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### General Rules

- Assign unique VMIDs per VM.
- Never reuse VMIDs without TLBI.
- Use generation-based VMID allocation for large systems.

### Nested Virtualization

- Use nested VMID ranges.
- Maintain shadow VMID tables.

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## 13.8 — Best Practices for Page Table Updates

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### Always follow BBM

1. Invalidate descriptor
2. DSB SY
3. TLBI
4. DSB SY
5. Install new descriptor
6. DSB SY
7. ISB

**Never update descriptors without barriers.**

**Never update descriptors without TLBI.**

**Never update descriptors in SMP systems without broadcast TLBI.**

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# 13.9 — Best Practices for TLB Invalidation

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### Correct TLBI Sequence

```
TLBI <operation>
DSB SY
ISB
```

### Rules

- Always invalidate by ASID when switching processes.
- Always invalidate by VMID when switching VMs.
- Always invalidate by VA when updating a single page.
- Always invalidate all entries when changing TCR/MAIR.

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# 13.10 — Best Practices for SMP Systems

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### Shareability

- Use Inner Shareable for all normal memory.
- Use Outer Shareable only for cluster-level coherence.

### Barriers

- Use DSB SY for global visibility.
- Use DMB SY for ordering.
- Use ISB after MMU configuration changes.

### TLBI

- Use broadcast TLBI instructions.
- Ensure all cores execute DSB SY.

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# 13.11 — Best Practices for Device Memory

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### Attributes

- Always use Device-nGnRnE.
- Never use cacheable attributes.
- Never use write-back or write-through.

### Ordering

- Use DSB SY after MMIO writes.
- Use DSB SY before MMIO reads.

### Alignment

- Ensure device accesses are aligned.
- Avoid unaligned atomic operations.

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# 13.12 — Best Practices for DMA

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### Before DMA Read

- Clean caches
- DSB SY

### After DMA Write

- Invalidate caches
- DSB SY

### Memory Attributes

- Use Normal WB for DMA buffers.
- Avoid Device memory for DMA buffers.

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# 13.13 — Best Practices for Virtualization

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### Stage-1

- Keep guest mappings simple.
- Avoid large numbers of small pages.
- Use blocks where possible.

### Stage-2

- Keep Stage-2 tables compact.
- Use blocks for large contiguous regions.
- Use pages for fine-grained VM isolation.

### Security

- Hypervisor must enforce non-exec pages.
- Hypervisor must enforce read-only pages.
- Hypervisor must override guest attributes.

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# 13.14 — Best Practices for QNX/RTOS

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### Deterministic Behavior

- Use static ASID ranges.
- Use static VMID ranges.
- Use deterministic TTBR1 mappings.
- Avoid dynamic remapping.

### Safety

- Enforce W^X.
- Use PXN/UXN aggressively.
- Use Device-nGnRnE for all MMIO.

### Partitioning

- Use separate TTBR0 regions per partition.
- Use separate ASID ranges per partition.

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# 13.15 — Best Practices for Memory Layout

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### Kernel

- Map kernel at high VA addresses.
- Use global mappings.
- Use blocks for kernel code/data.

### User Space

- Use pages for user memory.
- Use ASLR for security.
- Use separate TTBR0 per process.

### Devices

- Map devices at fixed VA/IPA/PA.
- Use Device-nGnRnE.
- Use pages for device registers.

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# 13.16 — Best Practices for Performance

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### Use Blocks

- Blocks reduce TLB pressure.
- Blocks reduce walk cache pressure.
- Blocks improve performance.

### Minimize Page Table Depth

- Use 48-bit VA.
- Avoid 52-bit VA unless necessary.

### Optimize TLB Behavior

- Keep kernel mappings global.
- Avoid frequent TTBR changes.
- Avoid excessive page fragmentation.

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# 13.17 — Best Practices for Security

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### W^X

- Never allow writable + executable pages.

### PXN/UXN

- Enforce non-exec for kernel data.
- Enforce non-exec for user data.

### PAN

- Prevent EL0 from accessing EL1 memory.

### Stage-2

- Hypervisor must enforce non-exec pages.
- Hypervisor must enforce read-only pages.

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## 13.18 — Best Practices for Reliability

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**Always use BBM.**

**Always use DSB SY + ISB.**

**Always use TLBI.**

**Always validate descriptor alignment.**

**Always validate TCR/MAIR configuration.**

**Always validate TTBR base addresses.**

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## 13.19 — Best Practices for Debuggability

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### Logging

- Log ESR_ELx, FAR_ELx, HPFAR_EL2.
- Log descriptor values.
- Log TTBR/TCR/MAIR values.

### Tracing

- Trace BBM sequences.
- Trace TLBI sequences.
- Trace ASID/VMID allocation.

### Diagnostics

- Provide tools to dump page tables.
- Provide tools to dump TLB entries.
- Provide tools to inspect walk cache behavior.

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# 13.20 — Chapter Summary

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This chapter covered:

- TTBR best practices
- TCR best practices
- MAIR best practices
- descriptor layout best practices
- ASID/VMID allocation

- page table update rules
- TLB invalidation rules
- SMP rules
- device memory rules
- DMA rules
- virtualization rules
- QNX/RTOS rules
- memory layout rules
- performance rules
- security rules
- reliability rules
- debugging rules